Harshil Bhatia

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EDUCATION

Carnegie Mellon University — *Pittsburgh, PA* | Dec 2026

Masters in Computer Vision — GPA: 4.22/4.0

Indian Institute of Technology, Jodhpur — *Jodhpur, India* | May 2023

Bachelors in Computer Science — GPA: 8.5/10

SKILLS

Languages & Frameworks: Python (pytorch, keras, numpy, pandas) C++, CUDA, Jupyter, Git, AWS, Docker, Linux, Conda.

Modeling & Architecture: Representation Learning, Deep Learning, Diffusion Models, LLMs, Autoregressive generation. Vision Language models, Transformers, Vision Transformers (ViTs), Video Models, World Models.

EXPERIENCE & RESEARCH PROJECTS

Frontier AI and Research - Amazon | *Applied Science Intern* — SF, USA | Jun 2026 - Present.

- Working on pretraining large robotics foundational models.

Avataar.ai | *Research Engineer-II* — Bangalore, India | Jun 2023 - Jun 2025.

- Led the development of a **Generative AI** system leveraging **Diffusion-based Foundation Models**, incorporating **Knowledge Distillation** principles for high-fidelity outputs.
- Co-created the **architecture** and feature extraction component of an **image-to-3D** pipeline, implemented with **multi-GPU** training, with mixed precision and checkpointing.
- Delivered real-time 3D reconstruction with **Gaussian Splatting** at 120+ FPS under 20 MB memory.
- Worked on **controllability** and **DPO** based alignment of generated images.
- Performed **Supervised Fine-Tuning (SFT)** of task-specific language models using **Low Rank Adaptation (LoRA)**.

RESEARCH PROJECTS

Generative Representation Learning for VLMs | *Carnegie Mellon University* — PA | March 2026 – Present.

Advisor: Dr. Max Simchowitz

- Post-training vision-language models with generative flow loss alongside perplexity, key insight is that the internal visual representations must be predictive and reconstructive rather than purely discriminative
- Motivated by the observation that captioning and VQA objectives don't enforce the model to learn how to use the visual representation properly and the language prior in the model takes over.

Instructable Tokenisation | *Carnegie Mellon University* — PA | Jan 2026 – March 2026.

Advisor: Dr. Max Simchowitz

- Developing **Instructable** and **flexible** tokenisers for Image and Robot Foundation Models, treating image encoding as a Supervised Fine-Tuning (SFT) problem guided by **Vision-Language Models** (VLMs).
- Trying to improve both the **generation efficiency** and the **discriminative ability** of the learnt representation.
- Developed a task conditioned router based approach to efficiently route and activate bottleneck registers.
- Applying DPO-style objectives to align token distributions with semantic intent, maximizing downstream generation quality as a reward signal for representation learning

3D Vision Language Action Models | *Carnegie Mellon University* — PA | Dec 2025 – Present.

Advisor: Dr. Shubham Tulsiani

- Developing end-to-end **Vision-Language-Action** models for robotic manipulation trained on 2D demonstrations with reinforcement learning-based reward shaping for policy optimization.
- Implemented learnable **RoPE-based positional encoding** for camera-robot spatial calibration.
- Extending the method to fine-tune SoTA VLAs like Groot and Pi on Droid-dataset.

SELECTED PUBLICATIONS

Summary: Published **8 peer-reviewed articles** with over **125 citations**, an h-index of 4, and an i10-index of 3 (Full Pub list).

[1] LiAutoFlow: Autoregressive Flow Matching for Continuous AV Scene Prediction in CVPRW, 2026 - **H. Bhatia**, S. Dasgupta

[2] CCuantuMM: Cycle-Consistent Quantum-Hybrid Matching of Multiple Shapes in **CVPR**, 2023 - **H. Bhatia**, E. Tretschk, Z. Lähner, M.S. Benkner, M. Moeller, C. Theobalt and V. Golyanik.

[3] CARFF: Conditional Auto-encoded Radiance Field for 3D Scene Forecasting in **ECCV**, 2024 - J. Yang, K. Desai, C. Packer, **H. Bhatia**, N. Rhinehart, R. McAllister, J. Gonzalez.

[4] SynthProv: Interpretable Framework for Profiling Identity Leakage in WACV, 2024 - J. Singh*, **H. Bhatia***, A. Bharati, R. Singh and M. Vatsa.